Geometry
Unit 9
Lesson 6 Practice

Name____________________
Date_________________
Period_________

1. Find the volume of a pyramid whose height is 12.9 feet and whose base is a rectangle with dimensions of 8.4 feet by 7.6 feet.

2. Find the volume of a rectangular prism with a width of 8.4 feet, a length of 7.6 feet, and a height of 12.9 feet.

3. What is the ratio of the answer from Question 1 to the answer from Question 2?

4. Find the volume of a sphere with a diameter of 2.82 meters. Use 3.14 for π. Round to the nearest hundredth.

5. Find the volume of a hemisphere with a diameter of 2.82 meters. Use 3.14 for π. Round to the nearest hundredth.

6. What is the relationship between the answers from Questions 4 and 5?

7. Determine the volume of a cone with a radius of 17.1 inches, a height of 6.9 inches, and a slant height of 18.44 inches in terms of π.

8. Find the volume of a cylinder with a radius of 28.3 inches and a height of 14.4 inches. Round your answer to the nearest thousandth.

9. A composite figure consists of a cylinder and a square pyramid. The height of the cylinder is 49.7 centimeters. The height of the pyramid is 26.5 centimeters. The base of the square pyramid has a width of 20 centimeters. The base of the pyramid sits on one base of the cylinder with the width of the square base equal to the diameter of the cylinder. Find the volume of the composite figure. Round to the nearest hundredth.

10. A composite figure consists of a cylinder and a hemisphere. The height of the cylinder is 21.4 inches. The base of the hemisphere sits on one base of the cylinder, both having a diameter of 16 inches. Find the volume of the composite figure. Use 3.14 for π. Round to the nearest hundredth.